

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

2. Motion Estimation Error Analysis

Two motion estimation techniques produce displacement errors (in pixels):

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1	5	8
R2	6	10
R3	5	9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object Stereo Vision Structure from Motion

A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

5. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional	85	100
Deep Learning	92	100

(a) Compute accuracy for both systems.

(b) Compare and evaluate which system performs better.

6. Precision-Recall Trade-off Analysis

Two models produce:

Model TP FP FN

A	70	30	20
B	85	25	25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced.

7. F1-Score Comparison

Two models:

Model Precision Recall

A	0.8	0.6
B	0.75	0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200
B	170	200

(a) Compute accuracy.

(b) Compare and evaluate.

10. Error Rate Comparison

Results:

Method	Errors	Total
A	20	100
B	10	100

(a) Compute error rate.

(b) Compare and evaluate.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Compute mean error.

(b) Compare variability and evaluate stability.

12. IOU Comparison

IOU values:

Object	A	B
1	0.6	0.8
2	0.7	0.85
3	0.65	0.9

(a) Compute average IOU.

(b) Compare and evaluate detection quality.

13. Recall Comparison in Detection

Results:

Model	TP	FN
A	60	40
B	80	20

(a) Compute recall.

(b) Compare and evaluate sensitivity.

14. Depth Estimation Error (Squared)

Predicted:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

(a) Compute squared error.

(b) Compare and evaluate accuracy.

15. Recognition Precision Comparison

Results:

System	TP	FP
A	90	10
B	85	5

(a) Compute precision.

(b) Compare and evaluate performance.

Code of Conduct:

I BHAVYA PS(192524017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Bhavya ps

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. Motion Estimation Variance

Given:

- Frame 1: Method A = 2, Method B = 6
- Frame 2: Method A = 3, Method B = 7
- Frame 3: Method A = 2, Method B = 5

Calculation:

Frame	Method A	Method B
1	2	6
2	3	7
3	2	5

Mean Error – Method A

$$\text{Mean A} = (2 + 3 + 2) / 3 = 7 / 3 = 2.33$$

Mean Error – Method B

$$\text{Mean B} = (6 + 7 + 5) / 3 = 18 / 3 = 6$$

Variance – Method A

$$\text{Variance A} = [(2-2.33)^2 + (3-2.33)^2 + (2-2.33)^2] / 3 \approx 0.22$$

Variance – Method B

$$\text{Variance B} = [(6-6)^2 + (7-6)^2 + (5-6)^2] / 3 = 0.67$$

Comparison:

- Method A has lower mean error (2.33) than Method B (6).
- Method A also has lower variance (0.22) than Method B (0.67).
- Lower variance indicates more consistent and stable errors.

Conclusion:

Method A is more stable and accurate because it has both lower mean error and lower variability. Method B has larger and more variable errors, making it less reliable for motion estimation.

2. IOU Comparison

Given:

- Object 1: A = 0.60, B = 0.80
- Object 2: A = 0.70, B = 0.85
- Object 3: A = 0.65, B = 0.90

Calculation:

Object	Method A	Method B
1	0.60	0.80
2	0.70	0.85
3	0.65	0.90

Average IOU – Method A

$$\text{Average IOU A} = (0.60 + 0.70 + 0.65) / 3 = 1.95 / 3 = 0.65$$

Average IOU – Method B

$$\text{Average IOU B} = (0.80 + 0.85 + 0.90) / 3 = 2.55 / 3 = 0.85$$

Comparison:

- Method A average IOU = 0.65.
- Method B average IOU = 0.85.
- A higher IOU means better overlap between predicted and actual bounding boxes.
- Method B has higher IOU for all three objects.

Conclusion:

Method B has better detection quality because its average IOU is 0.85 compared with 0.65 for Method A. Therefore, Method B provides more accurate object localization.

3. Recall Comparison in Detection

Given:

- Model A: TP = 60, FN = 40
- Model B: TP = 80, FN = 20

Calculation:

Model	TP	FN
A	60	40
B	80	20

Recall – Model A

$$\text{Recall A} = \text{TP} / (\text{TP} + \text{FN}) = 60 / (60 + 40) = 0.60 = 60\%$$

Recall – Model B

$$\text{Recall B} = \text{TP} / (\text{TP} + \text{FN}) = 80 / (80 + 20) = 0.80 = 80\%$$

Comparison:

- Model A recall = 60%.
- Model B recall = 80%.
- Higher recall means the system detects more of the actual positive objects.
- Model B has fewer missed detections because it has only 20 false negatives.

Conclusion:

Model B has better sensitivity and detection performance because its recall is 80%, compared with 60% for Model A. Therefore, Model B is preferable when minimizing missed detections is important.

4. Depth Estimation Error (Squared)

Given:

- Squared Error = (Predicted – Actual)²

Calculation:

Object	Predicted A	Predicted B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

Object 1

$$A: (2.5 - 2)^2 = 0.25 \quad B: (2.2 - 2)^2 = 0.04$$

Object 2

$$A: (3.5 - 3)^2 = 0.25 \quad B: (3.1 - 3)^2 = 0.01$$

Object 3

$$A: (4.5 - 4)^2 = 0.25 \quad B: (4.2 - 4)^2 = 0.04$$

Mean Squared Error – A

$$MSE A = (0.25 + 0.25 + 0.25) / 3 = 0.25$$

Mean Squared Error – B

$$MSE B = (0.04 + 0.01 + 0.04) / 3 = 0.09 / 3 = 0.03$$

Comparison:

- Method A MSE = 0.25.
- Method B MSE = 0.03.
- Lower squared error indicates that predicted depth is closer to actual depth.
- Method B has a much lower error than Method A.

Conclusion:

Method B is more accurate because its mean squared error is only 0.03, compared with 0.25 for Method A. Hence, Method B is preferable for accurate depth estimation.

5. Recognition Precision Comparison

Given:

- System A: TP = 90, FP = 10
- System B: TP = 85, FP = 5

Calculation:

System	TP	FP
A	90	10
B	85	5

Precision – System A

$$\text{Precision A} = \text{TP} / (\text{TP} + \text{FP}) = 90 / (90 + 10) = 0.90 = 90\%$$

Precision – System B

$$\text{Precision B} = \text{TP} / (\text{TP} + \text{FP}) = 85 / (85 + 5) = 85 / 90 \approx 0.9444 = 94.44\%$$

Comparison:

- System A precision = 90%.
- System B precision = 94.44%.
- Higher precision means fewer false positive detections.
- System B has fewer false positives (5) than System A (10).

Conclusion:

System B performs better in terms of precision, with 94.44% compared with 90% for System A. It is preferable when reducing false positives and improving recognition reliability are important.